

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third & Fourth Semester of B. Tech (EE) Examination

Apr-May 2018

EE201/EE201.01 Control Engineering

Date: 02.05.2018, Wednesday

Time: 10.00 a.m. To 01.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q-1 (a) Define the following terms [03]

1. Control
2. System
3. Control system

(b) Explain closed loop control system by giving suitable example. Also enlist advantages for the same. [07]

Q-2 (a) Obtain the transfer function using Mason's Gain formulae for the signal flow graph shown in **Figure I**. [05]

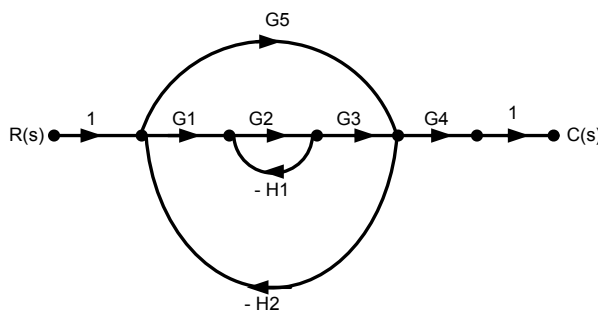


Figure – I

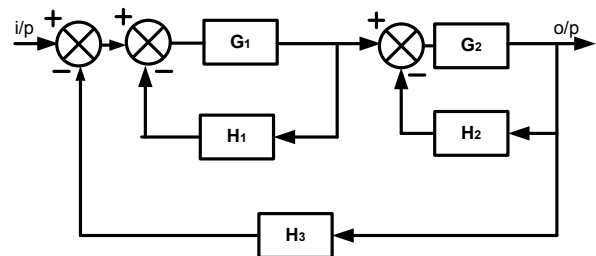


Figure - II

(b) Explain Missile launching system with the help of neat figure. [05]

OR

Q-2 (a) (i) Define following terms. [03]

Forward path, Source node, Sink node, Chain node, Feedback loop

(ii) Obtain T.F. for simple closed loop control system having forward path gain $G(s)$, Feedback path Gain $H(s)$. [02]

(b) Obtain Transfer Function using Block Diagram Reduction method for the block diagram shown in **Figure II**. [05]

Q-3 Attempt any three. [15]

(a) Explain the effects of PD controller in detail on 2nd order control system.

- (b) For a unity feedback system has $G(s) = \frac{40(s+2)}{s(s+1)(s+4)}$ Find
 (i) Type & Order of the system (ii) Error constants (iii) Error for ramp input
- (c) Explain in detail about various test signals used in control system. Also tabulate their values in Time domain and in Laplace domain.
- (d) Draw a typical response $c(t)$ with respect to time of 2nd Order system and Define all the terms with their mathematical equations..

SECTION – II

Q-4 Obtain the transfer function from the given circuit shown in **Figure III**. [04]

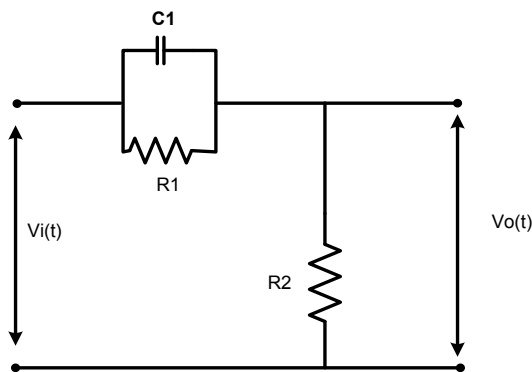


Figure – III

Q-5 (a) Draw the approximate root locus diagram for a closed loop control system whose open loop transfer function is given by [07]

$$G(s)H(s) = \frac{K}{s(s+5)(s+2)} \text{ . Comment on stability.}$$

- (b) The characteristics equation of the system is given by $s^6 + 4s^5 + 3s^4 - 16s^2 - 64s - 48 = 0$ [06]
 Find the number of roots of these equations with positive real part, zero real part and negative real part.

OR

Q-5 (a) Sketch the root locus of the system on the graph paper with open loop transfer [07]

$$\text{function of the system is given as } G(s)H(s) = \frac{K}{s(s+1)(s+2)(s+3)} \text{ .}$$

Where K is varying from zero to infinity.

- (b) The characteristics equation of the system is given by. [06]

I. $s^3 + 6s^2 + 11s + 6 = 0$

II. $s^3 + 4s^2 + s + 16 = 0$

Comment on stability using Routh - Hurwitz criterion.

Q-6 Attempt any three.**[18]**

- (a) A unity feedback system has

$$G(s) = \frac{2000}{s(s+1)(s+100)}.$$

Draw asymptotic Bode Plot. Determine Gain Margin.

- (b) A unity feedback system has

$$G(s) = \frac{10}{s(s+1)(s+2)}.$$
 Comment on stability using Nyquist Plot.

- (c) The Open Loop transfer function of unity Feed Back Control System is given by

$$G(s) = \frac{K}{s(1+0.001s)(1+0.25s)(1+0.1s)}.$$

Determine the value of K so that the system will have a phase margin of 40°

What will be the gain margin? Use Bode Plot.

- (d) Explain Phase Lead Compensation technique in detail.
